

Lecture#15

Broadcast Receiver

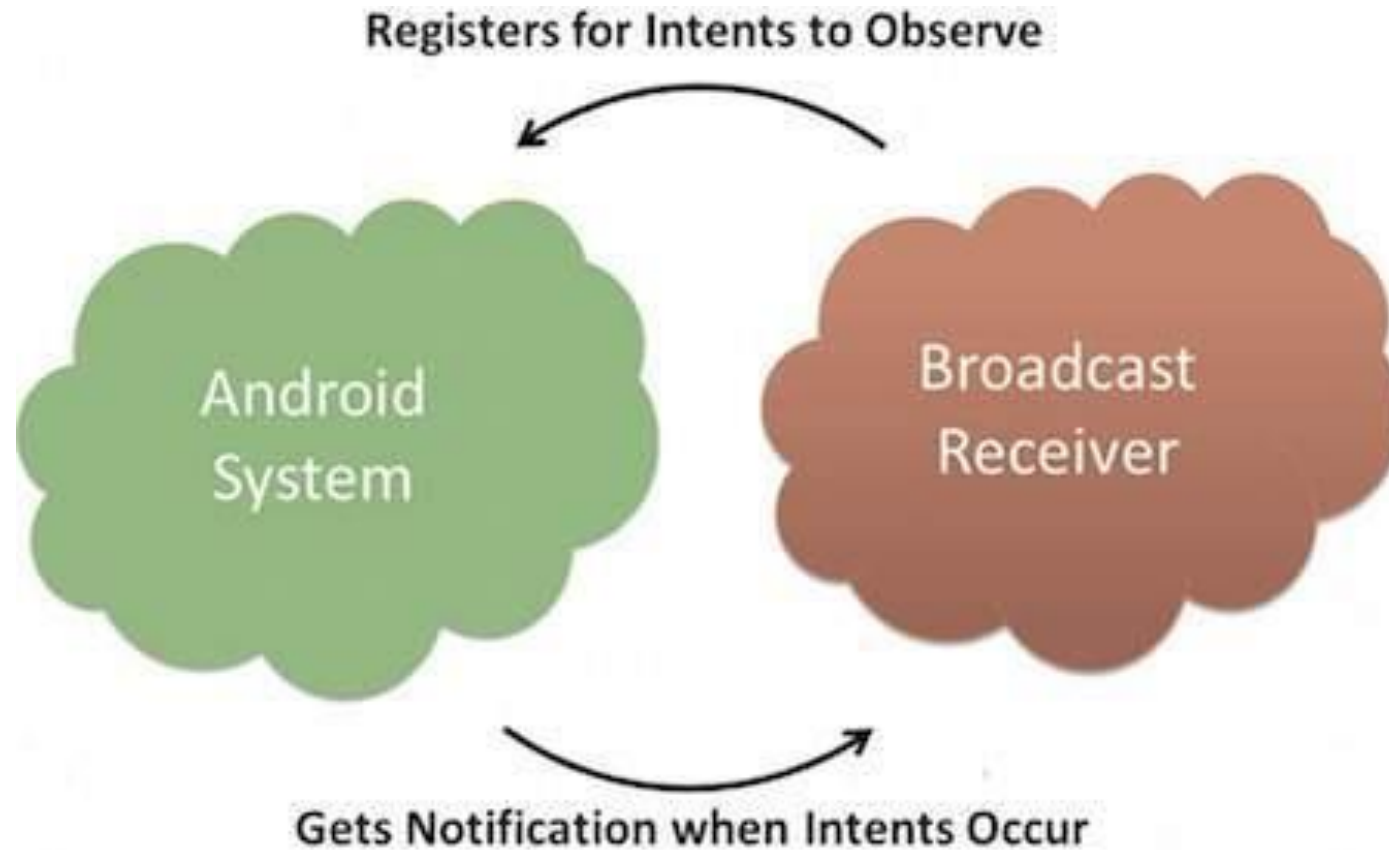
By

Dr. Muhammad Bilal

Broadcast Receivers

- Broadcast Receivers simply respond to broadcast messages from other applications or from the system itself
- These messages are sometime called events or intents
- For example, applications can also initiate broadcasts to let other applications know that some data has been downloaded to the device and is available for them to use, so this is broadcast receiver who will intercept this communication and will initiate appropriate action.

Broadcast Receiver Lifecycle



Steps Making Broadcast Receiver

- There are following two important steps to make **BroadcastReceiver** works for the system broadcasted intents
 - Creating the Broadcast Receiver
 - Registering Broadcast Receiver
- There is one additional steps in case you are going to implement your custom intents then you will have to create and broadcast those intents.

Creating the Broadcast Receiver

- A broadcast receiver is implemented as a subclass of **BroadcastReceiver** class and overriding the **onReceive()** method where each message is received as a **Intent** object parameter.

```
public class MyReceiver extends BroadcastReceiver {  
    @Override  
    public void onReceive(Context context, Intent intent) {  
        Toast.makeText(context, "Intent Detected: " + intent.getAction(), Toast.LENGTH_LONG).show();  
    }  
}
```

Registering Broadcast Receiver

- The broadcast receiver can be registered using

registerReceiver (BroadcastReceiver receiver, IntentFilter filter)

```
MyReceiver receiver = new MyReceiver();  
IntentFilter filter = new IntentFilter("android.intent.action.AIRPLANE_MODE");  
registerReceiver(receiver,filter);
```

MainBroadcastRecv.java

```
import android.content.IntentFilter;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import com.hitecuni.dce.lab4intent.R;

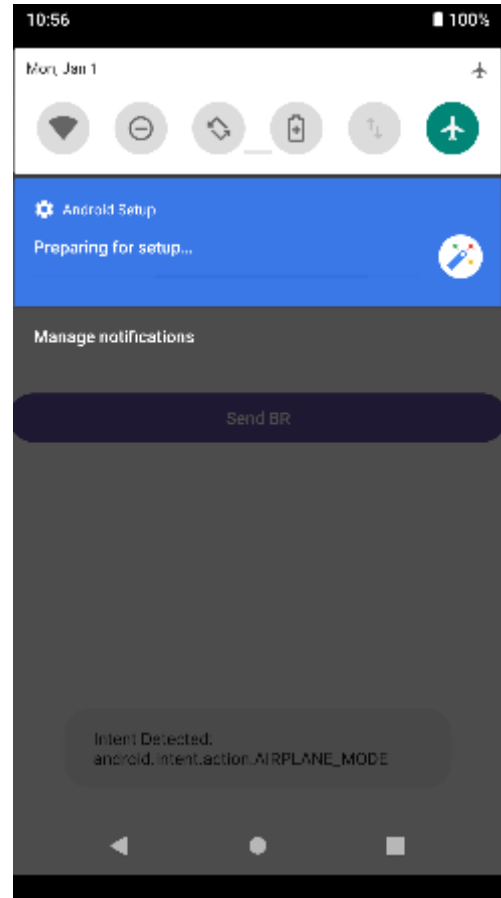
public class MainBroadcastRecv extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main_broadcast_recv);

        MyReceiver receiver = new MyReceiver();
        IntentFilter filter = new IntentFilter("android.intent.action.AIRPLANE_MODE");
        registerReceiver(receiver, filter);
    }
}
```

Emulator Results

Enable Airplane mode



More Standard Events

- There are several system generated events defined as final static fields in the Intent class. The following table lists a few important system events.

Sr.No	Event Constant & Description
1	android.intent.action.BATTERY_CHANGED Sticky broadcast containing the charging state, level, and other information about the battery.
2	android.intent.action.BATTERY_LOW Indicates low battery condition on the device.
3	android.intent.action.BATTERY_OKAY Indicates the battery is now okay after being low.
4	android.intent.action.BOOT_COMPLETED This is broadcast once, after the system has finished booting.
5	android.intent.action.CALL Perform a call to someone specified by the data.
6	android.intent.action.CALL_BUTTON The user pressed the "call" button to go to the dialer or other appropriate UI for placing a call.
7	android.intent.action.REBOOT Have the device reboot.

Broadcasting Custom Intents

- If you want your application itself should generate and send custom intents then you will have to create and send those intents by using the **sendBroadcast()** method inside your activity class.

```
public void broadcastIntent(View view) {  
    Intent intent = new Intent();  
    intent.setAction("com.example.mybroadcastreceiver.CUSTOM_INTENT");  
    sendBroadcast(intent);  
}
```

- This intent **com.example.mybroadcastreceiver.CUSTOM_INTENT** can also be registered in similar way as we have registered system generated intent.

```
MyReceiver receiver = new MyReceiver();  
IntentFilter filter = new IntentFilter("com.example.mybroadcastreceiver.CUSTOM_INTENT");  
registerReceiver(receiver,filter);
```

MainActivity.java

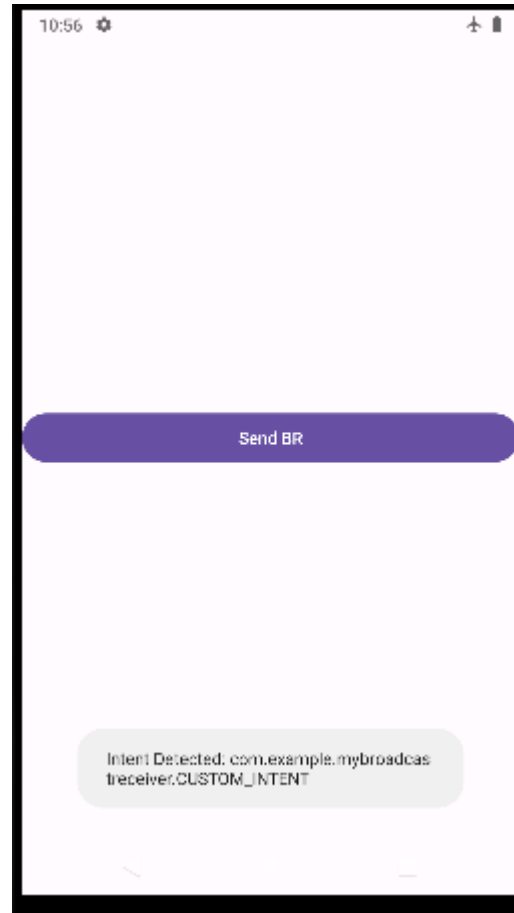
```
public class MainActivity extends AppCompatActivity {  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);  
    }  
  
    public void broadcastIntent(View view) {  
        Intent intent = new Intent();  
        intent.setAction("com.example.mybroadcastreceiver.CUSTOM_INTENT");  
        sendBroadcast(intent);  
    }  
}
```

MyReceiver.java

```
public class MyReceiver extends BroadcastReceiver {  
    @Override  
    public void onReceive(Context context, Intent intent) {  
        Toast.makeText(context, "Intent Detected: " + intent.getAction(), Toast.LENGTH_LONG).show();  
    }  
}
```

Emulator Results

On button press



by Dr. Muhammad Bilal